

Data Flow Diagrams (DFD)

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What is a DFD?

*Definition &
Core Concepts*

Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) is a graphical representation that shows how data flows through a system. It models the processes, data stores, external entities, and the flow of data between them — without showing control flow or timing.

Invented by:

Ed Yourdon & Tom DeMarco (1970s)

Also known as:

Process Model / Bubble Chart

Used in:

Structured System Analysis & Design

Main purpose:

Visualize data movement in a system

Key Components of a DFD

Four fundamental symbols that make up every DFD



External Entity

A person, organization, or system outside that sends or receives data.



Process

Transforms or manipulates data. Shown as a circle or rounded rectangle.



Data Store

Stores data for later use (e.g., database, file). Shown as open rectangle.



Data Flow

Arrow showing the direction in which data moves between components.

Types of DFD

DFDs are organized in hierarchical levels — each level provides a deeper view of the system

Level 0

Context Diagram

What:

Highest level. Shows the entire system as a single process with external entities and major data flows.

When:

First step in DFD creation. Provides a bird's-eye view.

1 process · External entities · No data stores

Level 1

Top-Level DFD

What:

Breaks the single process into major sub-processes, showing primary data stores and flows.

When:

After context diagram. Defines main functional areas.

Multiple processes · Data stores appear · Main flows

Level 2

Detailed DFD

What:

Decomposes each Level-1 process into smaller, more detailed sub-processes.

When:

Complex systems needing detailed design documentation.

Sub-processes · Detailed flows · Specific stores

Level 3+

Further Decomposition

What:

Further breaks down Level-2 processes. Rarely needed; most systems stop at Level 2.

When:

Very large/complex enterprise-level systems.

Deep detail · Used sparingly · Primitive processes

Why DFDs Matter

Importance in Software Engineering



Communication Tool

Provides a common visual language between developers, analysts, and non-technical stakeholders.



Requirement Analysis

Helps identify all inputs, outputs, and data transformations during requirements gathering.



System Design

Serves as a blueprint for structuring modules, databases, and processing logic.



Error Detection

Highlights missing data flows, redundant processes, or logical inconsistencies early in development.



Documentation

Acts as living documentation that evolves with the system and aids future maintenance.



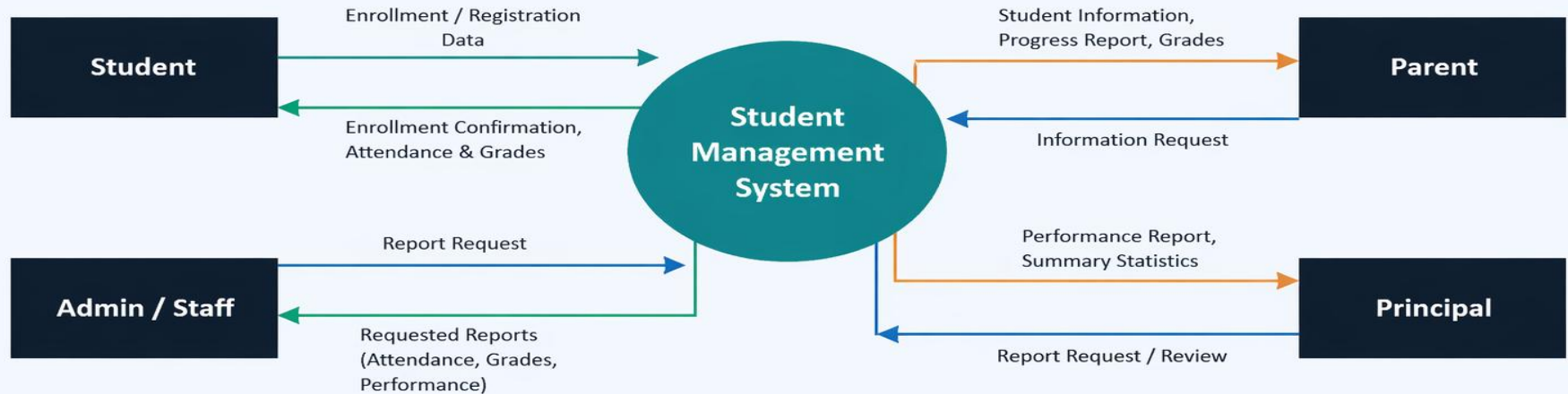
Validation & Verification

Used to verify that the system design matches the original requirements before coding begins.

Case Study 1: Student Management System

Context Diagram (Level 0) – Corrected

Note: This is a Context Diagram (Level 0). It shows the system as a single process and only external entities with data flows. Internal data stores are shown in Level 1 DFD.



Legend

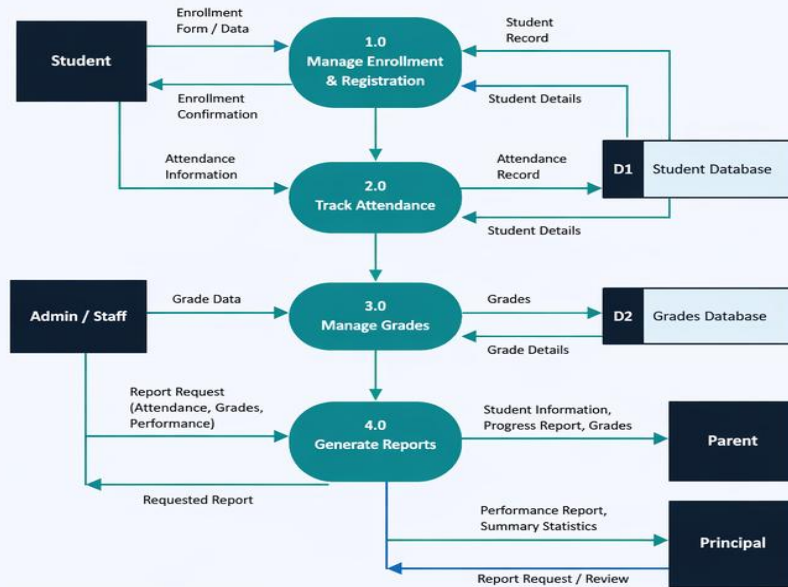
- External Entity
- Process
- Data Flow

CASE STUDY 1: STUDENT MANAGEMENT SYSTEM

DATA FLOW DIAGRAMS – LEVEL 1 AND LEVEL 2

LEVEL 1 DFD

Decomposition of the main process: Student Management System



Legend

External Entity

Process

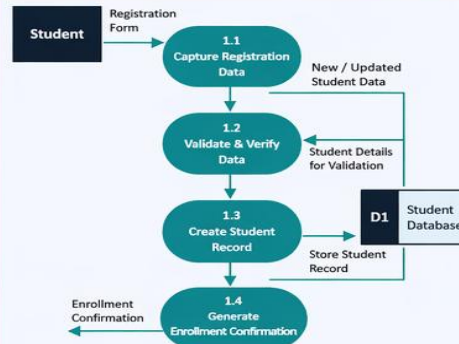
D1 Data Store

Level 1 Description

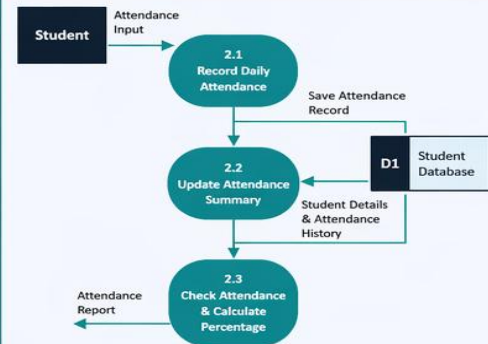
The main process is broken down into four subprocesses: Enrollment & Registration, Attendance Tracking, Grade Management, and Report Generation. Data stores are introduced to store student and grade data.

LEVEL 2 DFDs

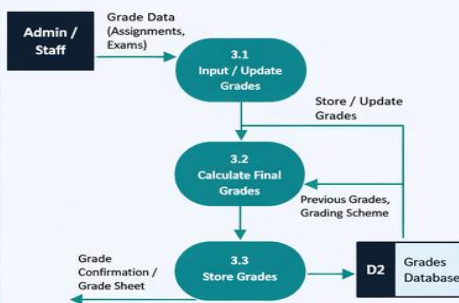
LEVEL 2 DFD - 1.0 Manage Enrollment & Registration



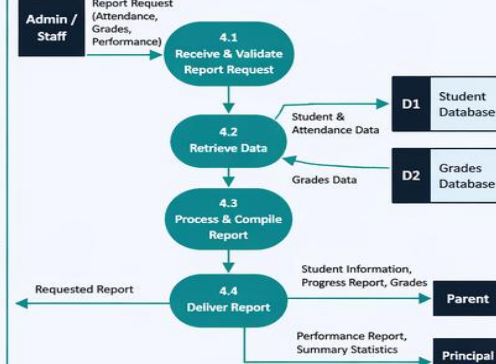
LEVEL 2 DFD - 2.0 Track Attendance



LEVEL 2 DFD - 3.0 Manage Grades

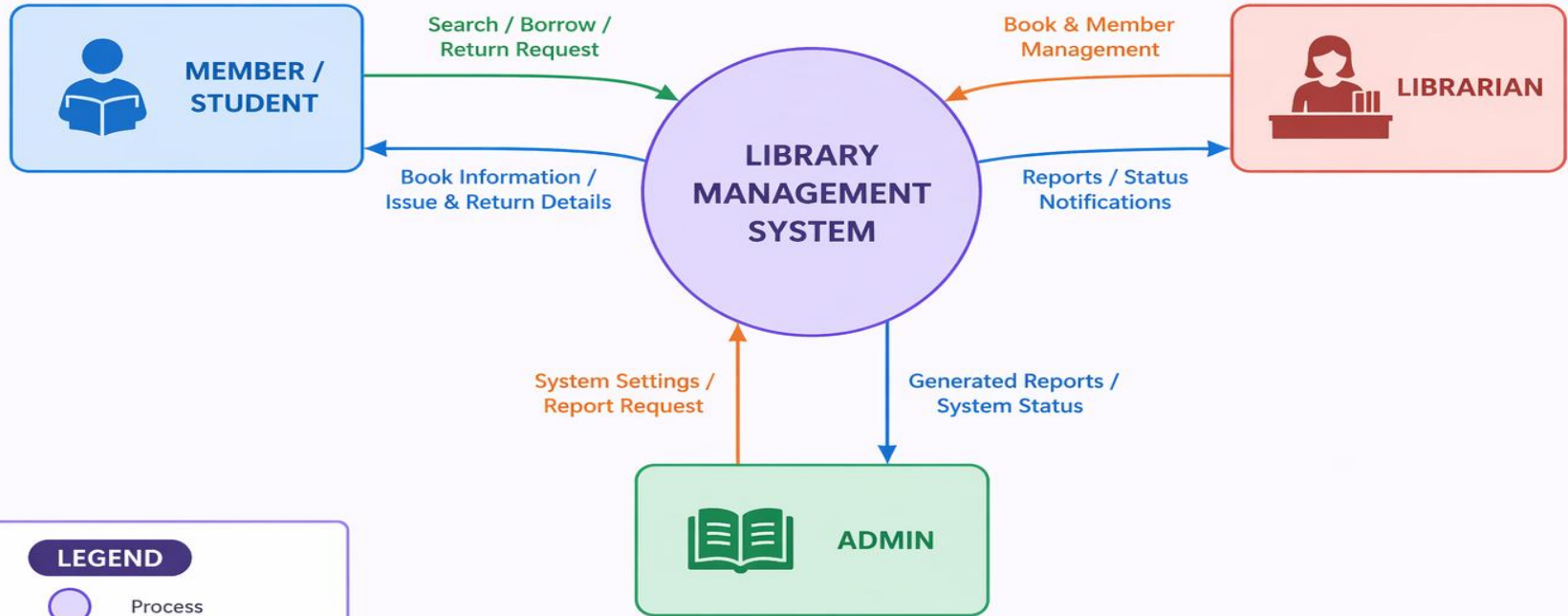


LEVEL 2 DFD - 4.0 Generate Reports



LIBRARY MANAGEMENT SYSTEM

LEVEL 0 DFD (Context Diagram)

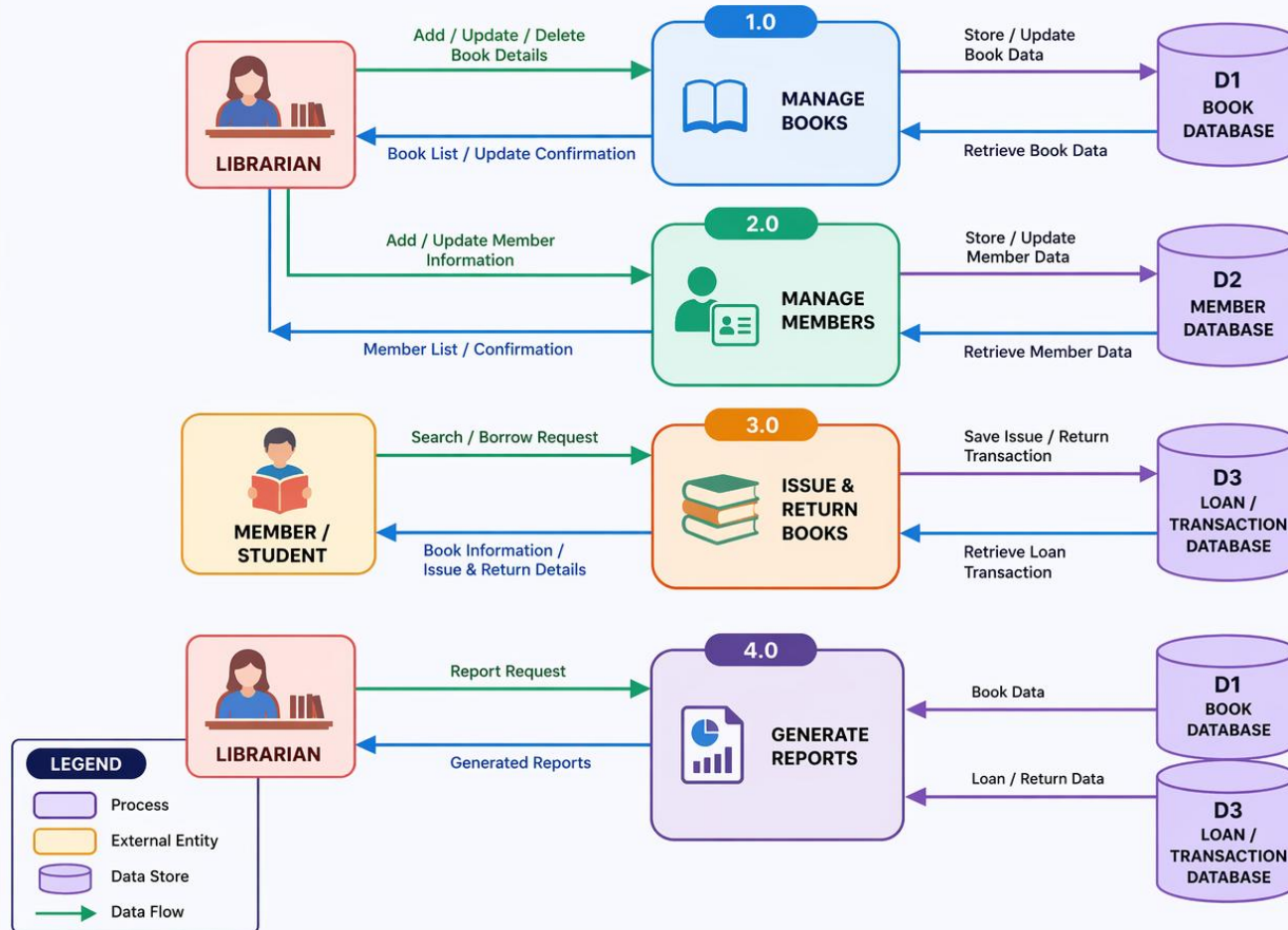


LEGEND

-  Process
-  External Entity
-  Data Flow

LEVEL 1 DFD – LIBRARY MANAGEMENT SYSTEM

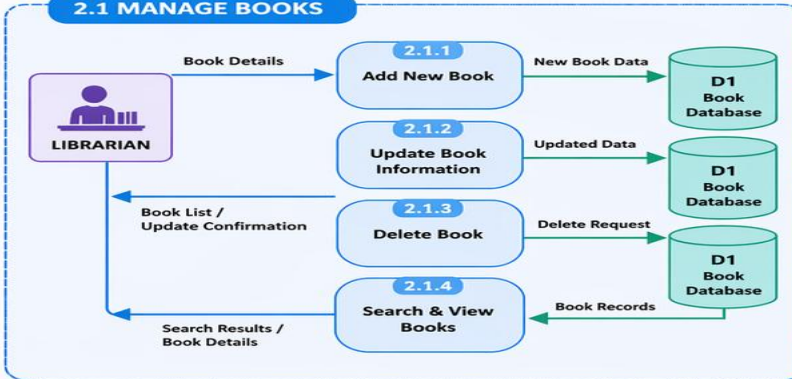
Decomposition of Library Management System (Level 0)



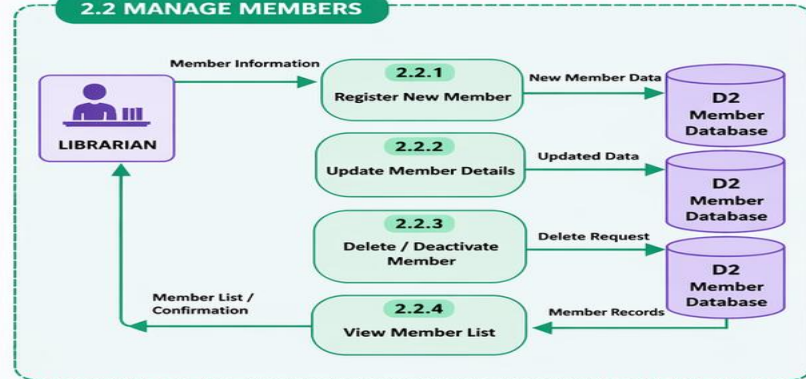
LEVEL 2 DFD – LIBRARY MANAGEMENT SYSTEM

Detailed Processes of Level 1

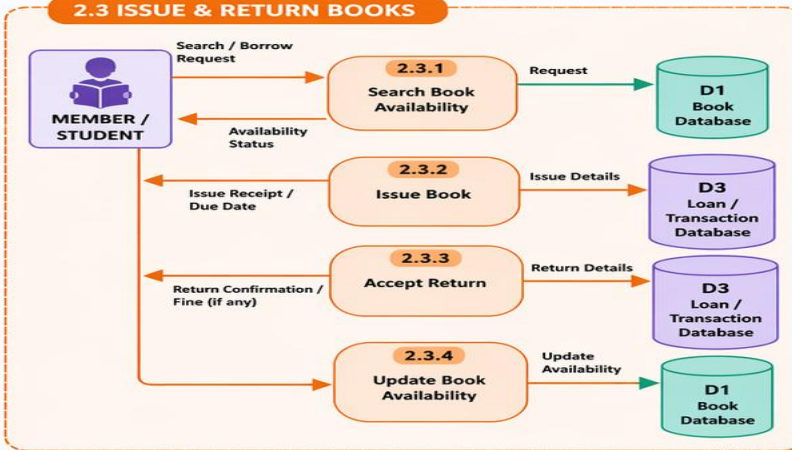
2.1 MANAGE BOOKS



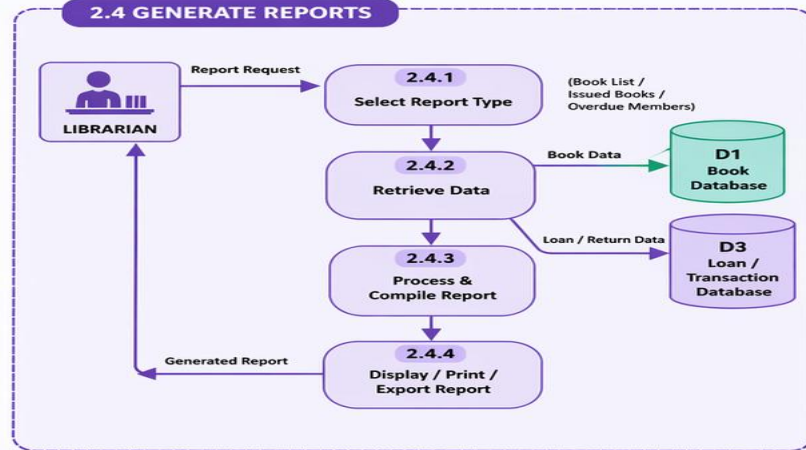
2.2 MANAGE MEMBERS



2.3 ISSUE & RETURN BOOKS



2.4 GENERATE REPORTS



LEGEND



External Entity



Process



Book Database (Data Store)



Member / Loan Database (Data Store)



Data Flow

DFD Rules & Best Practices



Rules to Follow

- All processes must have at least one input and one output
- Data stores must be connected to at least one process
- External entities cannot communicate directly — always via a process
- Label every data flow with a meaningful name
- Each process should be uniquely numbered
- Avoid crossing data flow lines when possible
- Balance DFD levels — inputs/outputs must be consistent



Common Mistakes

- X Black holes: processes with only inputs and no outputs
- X Miracles: processes with only outputs and no inputs
- X Grey holes: outputs smaller in scope than inputs
- X Connecting two external entities directly
- X Connecting two data stores without a process
- X Unlabeled or vaguely labeled data flows
- X Skipping levels in the hierarchy (jumping Level 0 → Level 2)

DFD vs. Flowchart — What's the Difference?

Aspect	Data Flow Diagram	Flowchart
Focus	Data movement in a system	Sequence of steps/actions
Shows Control?	No decision/loop logic	Decisions, loops, conditionals
Level of Detail	System-level abstraction	Step-by-step process detail
Primary Use	System analysis & design	Algorithm & program logic design
Audience	Analysts, stakeholders, developers	Developers, programmers
Symbols Used	Rectangle, Oval, Arrow, Open Box	Terminator, Process, Decision, Flow
Timing Shown?	No, static data model	Yes, sequential execution order

Thank You

